

Software Requirements Specification (SRS)

Project: jom-QA Automated Test Engine

This document outlines the core functional requirements for the jom-QA platform MVP.

Module 1: Authentication & Connection

FR-001: Secure User Login

The system must allow registered users to authenticate securely using an email and password. Upon successful authentication, the system must generate a secure JWT token and redirect the user to the main dashboard screen.

FR-002: Flutter VM Service Connection

The local runner must accept a valid WebSocket URI pointing to a running Flutter application's VM Service. The system must validate the connection handshake before enabling any live runtime interactions.

Module 2: Test Orchestration

FR-003: Live UI Inspection

The core engine must query the running Flutter app via Marionette to retrieve a list of interactive UI elements. Layout-only widgets like Paddings or Spacers must be filtered out locally to minimize token overhead.

FR-004: Autonomous Action Execution

Based on the AI model decision, the orchestrator must command the remote app to simulate user gestures including tap actions, text inputs into text fields, and scrolling routines.

Module 3: Automated Bug Reporting

FR-005: Trello Ticket Generation

When an unexpected error or assertion failure is captured during a test run, the system must automatically create a new card in the user's configured Trello board under the 'Bug Reports' list.

FR-006: GitHub Issue Dispatch

The system must format error logs and failure context into Markdown syntax and automatically open a new issue in the specified GitHub repository containing the requirement traceability tracking tags.